

Supplementary theory and analyses for “Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China”

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This supplement provides the proof of Proposition 1, a process-level strong-mixing result, additional applied analyses and the simulation specifications and results. The applied tables use 33 evaluation summers; 2015 and 2022 were reserved for method development and are excluded from these summaries.

Analytical roles

Table S1 uses three status families: primary specification or primary finite-record estimate, exploratory analysis, and pre-access extension. Descriptors within a family identify inferential, structural, reference or sensitivity purposes. Unless the table identifies an alternative estimand, every analysis uses the original 121-site labels, five bandwidths and equal-month, equal-scale and equal-summer estimator. Exploratory analyses were undertaken after completion of the primary analysis. The historical and station-extension plans were finalised after the primary analysis and before the extension data were retrieved. They assess temporal transfer and measurement agreement under the original event definition.

Table S1: Analytical status and purpose of the applied analyses.

Element	Analytical status	Data or support	Scientific purpose
121-site support, UTC daily peak field and quality rules	Primary specification	2015 and 2022 development summers	Data construction for the 33 evaluation summers
Within-record quartile labels, five Gaussian bandwidths, raw ratio and equal aggregation	Primary specification	Development summers and outcome-free simulations	Definition of the finite-record estimand
33-summer evaluation-period contrast	Primary finite-record estimate	33 evaluation summers	Application of the fixed specification to the evaluation record

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Table S1 continued

Element	Analytical status	Data or support	Scientific purpose
Student, calendar-lag HAC and sign summaries	Exploratory reference analysis	33 evaluation summers and outcome-free summer simulations	Descriptive and process-model comparisons
Global 99,999-draw product shift	Exploratory inferential analysis	33 evaluation summers	Conditional randomisation assessment under the product-invariance null for the observed records and fixed statistic
Physical Q_h /RMS reporting and alternative effect measures	Exploratory sensitivity analysis	33 evaluation summers	Expression of the result in $^{\circ}\text{C}^2$, $^{\circ}\text{C}$ and alternative ratio scales
Anomaly fields, latitude and planar bases, and climatology–anomaly energy	Exploratory structural analysis	33 evaluation summers	Decomposition of the changing spatial component
Historical climatology–anomaly decomposition	Exploratory structural analysis	41 historical summers	Period-specific accounting of anomaly energy and climatology–anomaly alignment
Elevation-augmented spatial basis	Exploratory structural analysis	33 evaluation summers and invariant geopotential	Assessment of the broad structured component after adding elevation to the geographic basis
Nested 465-site support	Exploratory sensitivity analysis	Four nested or partially nested supports	Spatial-support sensitivity with the original event times and labels
31-point bandwidth curve and four-configuration convergence comparison	Exploratory sensitivity analysis	121-site and nested-grid outcomes	Scale-profile smoothness and discretisation sensitivity
1950–1990 ERA5-Land extension	Pre-access extension	41 JJA summers	Temporally separated evaluation using the same reanalysis product and fixed 121-site estimator
Historical 99,999-draw product shift	Pre-access extension	123 historical month-year records	Protocol-specified conditional randomisation assessment of the fixed historical-extension statistic
Continuous intensity and $\pm 3/\pm 5$ -day matching	Exploratory sensitivity analysis	33 evaluation summers	Sensitivity to quartile categorisation and seasonal calendar separation
Heavy-tailed ratio comparison	Exploratory simulation	Heavy-tailed stress design	Comparison of raw, log and bounded statistics
Cross-record dependence simulation	Exploratory simulation	Simulated two-summer records	Sensitivity of product-shift size to shared temporal dependence across monthly records

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Table S1 continued

Element	Analytical status	Data or support	Scientific purpose
Ten-year NOAA-ERA5-Land extension	Pre-access extension	Ten evaluation years	External measurement and temporally separated effect comparison

S1 Finite-record product cyclic-shift validity

Let X collect the fixed regional-mean records used to define the event labels, and let D collect the corresponding daily graph-dispersion profiles. The finite group G acts on D by independently rotating the complete multiscale profile within each month-year record. Conditional on X , suppose

$$(D \mid X) \stackrel{d}{=} (gD \mid X) \quad \text{for every } g \in G. \quad (\text{S1})$$

For a measurable lower-tail statistic $T(X, D)$ fixed before the transformations are examined, define

$$p_G(X, D) = \frac{1}{|G|} \sum_{g \in G} I\{T(X, gD) \leq T(X, D)\}.$$

Proposition 1 (Product cyclic-shift validity). *For every $u \in [0, 1]$,*

$$\Pr\{p_G(X, D) \leq u \mid X\} \leq u \quad \text{almost surely.}$$

If the $|G|$ transformed statistics have no ties, p_G is uniform on $\{1/|G|, \dots, 1\}$ conditional on the invariant orbit. If $H_1, \dots, H_B$ are independent uniforms on G , independently of (X, D) , then

$$p_B = \frac{1 + \sum_{b=1}^B I\{T(X, H_b D) \leq T(X, D)\}}{B + 1}$$

satisfies

$$\Pr\{p_B \leq u \mid X\} \leq \frac{\lfloor (B + 1)u \rfloor}{B + 1} \leq u.$$

For fixed scale statistics $T_1, \dots, T_H$, put $M(X, D) = \min_h T_h(X, D)$ and define

$$p_h^{\text{adj}} = \frac{1}{|G|} \sum_{g \in G} I\{M(X, gD) \leq T_h(X, D)\}.$$

Under the complete joint invariance null, $\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u$. This is weak family-wise control; the proposition does not give strong control under arbitrary partial nulls.

Proof. Fix a value x in the probability-one set on which the conditional invariance in Equation (S1) holds, and write $S(y) = T(x, y)$. Let $m = |G|$.

For fixed y , form the multiset $\{S(gy) : g \in G\}$. For each $g \in G$, let

$$r_g(y) = \sum_{h \in G} I\{S(hy) \leq S(gy)\}.$$

Right multiplication by g permutes the elements of G . Hence

$$\begin{aligned} p_G(x, gy) &= \frac{1}{m} \sum_{h \in G} I\{S(hgy) \leq S(gy)\} \\ &= \frac{r_g(y)}{m}. \end{aligned}$$

If U is uniform on G , independently of y , then

$$\Pr\{p_G(x, Uy) \leq u \mid y\} = \frac{1}{m} \#\{g : r_g(y) \leq mu\}. \quad (\text{S2})$$

Order the m values in the multiset. An observation at the upper end of a tie block has weak rank equal to the final position of that block. Every tie block counted on the right of Equation (S2) therefore ends no later than position $\lfloor mu \rfloor$. The number of counted group elements is at most $\lfloor mu \rfloor$, and

$$\Pr\{p_G(x, Uy) \leq u \mid y\} \leq \frac{\lfloor mu \rfloor}{m} \leq u.$$

This inequality also holds when the action has a nontrivial stabiliser. Repeated transformations then appear as ties in the multiset and make the weak-rank test conservative.

Let U now be independent of (X, D) and uniform on G . Conditional invariance gives $UD \mid X = x \stackrel{d}{=} D \mid X = x$. Applying the preceding inequality conditionally on D yields

$$\begin{aligned} \Pr\{p_G(x, D) \leq u \mid X = x\} &= \Pr\{p_G(x, UD) \leq u \mid X = x\} \\ &= \mathbb{E}[\Pr\{p_G(x, UD) \leq u \mid X = x, D\} \mid X = x] \\ &\leq u. \end{aligned}$$

If the transformed statistics have no ties, the ranks $r_g(y)$ are $1, \dots, m$ once each. Equality holds at $u = j/m$, proving uniformity on the attainable grid.

For the sampled version, take $U_0, U_1, \dots, U_B$ independently and uniformly from G , independently of (X, D) . Conditional on (X, D) , the variables $S(U_0D), \dots, S(U_BD)$ are identically distributed and exchangeable. The weak rank of the first among these $B + 1$ values satisfies

$$\Pr \left[\frac{1 + \sum_{b=1}^B I\{S(U_bD) \leq S(U_0D)\}}{B + 1} \leq u \mid X = x \right] \leq \frac{\lfloor (B + 1)u \rfloor}{B + 1}. \quad (\text{S3})$$

It remains to relate the random anchor to the implemented construction, whose first transformation is the identity. Put $H_b = U_b U_0^{-1}$ for $b = 1, \dots, B$ and $D^* = U_0 D$. The map

$$(U_0, U_1, \dots, U_B) \mapsto (U_0, H_1, \dots, H_B)$$

is a bijection on G^{B+1} . Thus $H_1, \dots, H_B$ are independent uniforms and are independent of U_0 . Moreover, $D^* \mid X = x \stackrel{d}{=} D \mid X = x$. The vector in Equation (S3) consequently has the same conditional distribution as

$$\{S(D), S(H_1D), \dots, S(H_BD)\}.$$

This is the identity-plus- B construction used to calculate p_B .

The statistic must remain a fixed measurable function of the jointly shifted outcome matrix. For the scale-wise single-step comparison, let T_h denote the scale statistics and put $M(y) = \min_h T_h(x, y)$. The adjusted value

$$p_h^{\text{adj}} = \frac{1}{m} \sum_{g \in G} I\{M(gy) \leq T_h(x, y)\}$$

is nondecreasing in its observed threshold. Therefore $\min_h p_h^{\text{adj}}$ is attained at a scale for which $T_h(x, y) = M(y)$ and equals the randomisation value for M . Applying the enumeration result to M gives

$$\Pr\{\min_h p_h^{\text{adj}} \leq u \mid X\} \leq u.$$

The conclusion uses the complete joint null because M combines every scale. This completes the proof. $\square$

S2 Process-level strong-mixing and HAC reference

Let $\widehat{\mathbf{R}}_r = (\widehat{R}_{h_1 r}, \dots, \widehat{R}_{h_H r})^\top$ be the scale profile for summer r , $\mathbf{a} = H^{-1} \mathbf{1}_H$, $Z_r = \mathbf{a}^\top \widehat{\mathbf{R}}_r$ and $\widehat{\theta}_R = R^{-1} \sum_{r=1}^R Z_r$. Write $\gamma(k) = \text{Cov}(Z_0, Z_k)$ and define

$$\begin{aligned} \widehat{\gamma}_R(k) &= \frac{1}{R} \sum_{r=k+1}^R (Z_r - \overline{Z}_R)(Z_{r-k} - \overline{Z}_R), \\ \widehat{\sigma}_R^2 &= \widehat{\gamma}_R(0) + 2 \sum_{k=1}^{L_R} K\left(\frac{k}{L_R + 1}\right) \widehat{\gamma}_R(k). \end{aligned} \quad (\text{S4})$$

Theorem S1 (Strong-mixing limit and HAC reference). *Suppose $\{\widehat{\mathbf{R}}_r\}_{r \in \mathbb{Z}}$ is strictly stationary and strongly mixing, with $\mathbb{E}\|\widehat{\mathbf{R}}_0\|^{2+\delta} < \infty$ for some $\delta > 0$ and $\sum_{k=1}^{\infty} \alpha(k)^{\delta/(2+\delta)} < \infty$. Then every entry of*

$$\boldsymbol{\Omega} = \sum_{k=-\infty}^{\infty} \text{Cov}(\widehat{\mathbf{R}}_0, \widehat{\mathbf{R}}_k)$$

is absolutely convergent. If $\sigma^2 = \mathbf{a}^\top \boldsymbol{\Omega} \mathbf{a} > 0$ and $\theta = \mathbf{a}^\top \mathbb{E}(\widehat{\mathbf{R}}_0)$, then

$$\sqrt{R}(\widehat{\theta}_R - \theta) \xrightarrow{d} N(0, \sigma^2).$$

Suppose in addition that K is bounded on $[0, \infty)$, $K(0) = 1$, is continuous at zero and vanishes outside $[0, 1]$; $L_R \rightarrow \infty$, $L_R/R \rightarrow 0$; and

$$\sum_{k=0}^{L_R} |\widehat{\gamma}_R(k) - \gamma(k)| = o_p(1). \quad (\text{S5})$$

Then $\widehat{\sigma}_R^2 \xrightarrow{p} \sigma^2$ and

$$\frac{\sqrt{R}(\widehat{\theta}_R - \theta)}{\widehat{\sigma}_R} \xrightarrow{d} N(0, 1).$$

For a ratio component $\widehat{R}_{jr} = A_{jr}/B_{jr} - 1$, with $A_{jr} \geq 0$ and $B_{jr} > 0$, a sufficient condition for its $(2 + \delta)$ moment is the existence of conjugate $p, q > 1$ such that

$$\mathbb{E} A_{j0}^{(2+\delta)p} < \infty, \quad \mathbb{E} B_{j0}^{-(2+\delta)q} < \infty.$$

The same conclusion holds for a finite average of such ratios when the condition holds for every constituent ratio.

Proof. Let $\boldsymbol{\mu} = \mathbb{E}(\widehat{\mathbf{R}}_0)$. A measurable transformation cannot increase strong-mixing coefficients, so $\{Z_r\}$ is strictly stationary and strongly mixing with coefficients bounded by $\alpha(k)$. The inequality $|\mathbf{a}^\top v| \leq \|\mathbf{a}\| \|v\|$ gives $\mathbb{E} |Z_0|^{2+\delta} < \infty$.

Put $\xi_r = Z_r - \mathbb{E} Z_0$. Davydov's covariance inequality, with both moment exponents equal to $2 + \delta$, gives a finite constant C such that

$$|\gamma(k)| = |\mathbb{E}(\xi_0 \xi_k)| \leq C \|\xi_0\|_{2+\delta}^2 \alpha(|k|)^{\delta/(2+\delta)}, \quad k \neq 0. \quad (\text{S6})$$

The mixing-rate assumption implies $\sum_{k=-\infty}^{\infty} |\gamma(k)| < \infty$. Applying the same inequality componentwise establishes absolute convergence of every entry of $\boldsymbol{\Omega}$. Bilinearity and absolute convergence permit interchange of the finite quadratic form and the infinite sum:

$$\mathbf{a}^\top \boldsymbol{\Omega} \mathbf{a} = \sum_{k=-\infty}^{\infty} \text{Cov}(Z_0, Z_k) = \sum_{k=-\infty}^{\infty} \gamma(k) = \sigma^2. \quad (\text{S7})$$

The central limit theorem for strongly mixing sequences applies under the stated moment and mixing-rate conditions ([Ibragimov and Linnik, 1971](#)). Hence

$$R^{-1/2} \sum_{r=1}^R \xi_r \xrightarrow{d} N(0, \sigma^2).$$

Since $\mathbb{E} Z_0 = \mathbf{a}^\top \boldsymbol{\mu} = \theta$ and $R^{-1} \sum_r Z_r = \widehat{\theta}_R$, this proves the first limit.

For consistency of Equation (S4), define

$$\sigma_{K, L_R}^2 = \gamma(0) + 2 \sum_{k=1}^{L_R} K \left(\frac{k}{L_R + 1} \right) \gamma(k).$$

Boundedness of K and Equation (S5) give

$$\begin{aligned} |\widehat{\sigma}_R^2 - \sigma_{K, L_R}^2| &\leq |\widehat{\gamma}_R(0) - \gamma(0)| \\ &\quad + 2 \|K\|_\infty \sum_{k=1}^{L_R} |\widehat{\gamma}_R(k) - \gamma(k)| = o_p(1). \end{aligned}$$

For each fixed k , continuity at zero and $L_R \rightarrow \infty$ imply $K\{k/(L_R + 1)\} \rightarrow 1$. The weights are uniformly bounded, vanish for $k > L_R$, and $\sum_k |\gamma(k)| < \infty$. Dominated convergence for series therefore yields

$$\sigma_{K, L_R}^2 \longrightarrow \gamma(0) + 2 \sum_{k=1}^{\infty} \gamma(k) = \sigma^2.$$

Thus $\hat{\sigma}_R^2 \xrightarrow{p} \sigma^2$. Positivity of σ^2 and Slutsky's theorem give the studentised limit.

Condition (S5) is separate from the assumptions used for the central limit theorem. Primitive sufficient conditions for it require higher moments, stronger mixing-rate summability and a compatible bandwidth rate (Andrews, 1991; Newey and West, 1987).

For comparison, fix a lag L . Strong mixing implies ergodicity. The ergodic theorem applied to $(Z_r - \mathbb{E} Z_0)(Z_{r-k} - \mathbb{E} Z_0)$, together with $\bar{Z}_R \xrightarrow{p} \mathbb{E} Z_0$, gives $\hat{\gamma}_R(k) \xrightarrow{p} \gamma(k)$ for each fixed k . The fixed-lag estimator therefore converges to

$$\sigma_{K,L}^2 = \gamma(0) + 2 \sum_{k=1}^L K \left(\frac{k}{L+1} \right) \gamma(k). \quad (\text{S8})$$

For Bartlett lag $L = 2$, this is $\gamma(0) + 2\{(2/3)\gamma(1) + (1/3)\gamma(2)\}$. Its difference from the full long-run variance is

$$\sigma^2 - \sigma_{K,2}^2 = 2 \left\{ \frac{1}{3}\gamma(1) + \frac{2}{3}\gamma(2) + \sum_{k=3}^{\infty} \gamma(k) \right\}.$$

The fixed-lag estimator is consistent only if this weighted error is zero.

It remains to verify the stated sufficient ratio condition. With $s = 2 + \delta$, convexity and Hölder's inequality give

$$\begin{aligned} \mathbb{E} |A_{j0}/B_{j0} - 1|^s &\leq 2^{s-1} \left\{ \mathbb{E}(A_{j0}^s B_{j0}^{-s}) + 1 \right\} \\ &\leq 2^{s-1} \left[\{\mathbb{E} A_{j0}^{sp}\}^{1/p} \{\mathbb{E} B_{j0}^{-sq}\}^{1/q} + 1 \right] < \infty. \end{aligned}$$

A finite sum of variables with finite s th moments also has a finite s th moment. Applying this argument to each component proves the final claim. $\square$

The observed middle-day denominators, 495 values across 99 records and five scales, range from 2.25 to 25.99 °C². This range excludes a numerical near-zero denominator in the observed record but does not establish the negative-moment condition required by the process-level ratio argument. The process theorem and finite-record product-shift analysis therefore address distinct inferential settings.

S3 Global product-shift result

The exploratory product-shift analysis used the statistic $T = \hat{\theta}_{y_H}$. Each draw independently selected one cyclic offset in each of the 99 evaluation month-year records and shifted all five graph-dispersion columns together. Regional-mean series and event labels remained in their observed order. The statistic was then reconstructed with equal months, scales and summers. The generator seed was 20260809 and $B = 99,999$.

No sampled statistic was at or below the observed -7.2834% . The plus-one value is therefore $1/(99,999 + 1) = 0.00001$, its minimum attainable value for this run. The sampled null had mean 0.8241% , standard deviation 1.5651 points and median 0.8139% ; its 0.1% , 1%

and 5% quantiles were -3.8788% , -2.7440% and -1.7225% . A ratio-based randomisation distribution need not be centred at zero, and the test compares the observed statistic with that distribution directly.

Table S2: The 99,999-draw product cyclic-shift analysis. Effects, null means and null standard deviations are in percentage points. The last column uses the minimum statistic over the five scales and has weak family-wise validity only under the complete joint invariance null.

Bandwidth (km)	Observed (points)	Null mean (points)	Null SD (points)	Unadjusted p	Complete-null p
126	-2.856	0.359	1.059	0.00104	0.02504
252	-3.945	0.531	1.301	0.00022	0.00571
503	-5.907	0.859	1.683	0.00002	0.00019
1,006	-10.441	1.125	1.964	0.00001	0.00001
2,013	-13.268	1.246	2.089	0.00001	0.00001
Five-scale mean	-7.283	0.824	1.565	0.00001	—

The global row is an exploratory finite-record test formulated after the initial 33-summer estimate. The statistic was fixed before evaluation, whereas the decision to calculate its randomisation distribution followed that initial estimate. Its null is the conditional product-invariance statement in Equation (S1); a stationary long-run climate parameter lies outside the scope of this test. The scale-wise rows summarize the same joint draws and are mutually dependent.

S4 Physical scale of the graph contrast

For a field $\mathbf{y}$,

$$Q_h(\mathbf{y}) = \frac{\sum_{i<j} w_{ij,h} (y_i - y_j)^2}{2 \sum_{i<j} w_{ij,h}}, \quad D_h^{\text{RMS}}(\mathbf{y}) = \{2Q_h(\mathbf{y})\}^{1/2}.$$

Thus Q_h has units $^\circ\text{C}^2$ and D_h^{RMS} is the weighted root mean squared pairwise difference in $^\circ\text{C}$. Table S3 reports equal-month and equal-summer means of Q_h and applies $D_h^{\text{RMS}} = (2Q_h)^{1/2}$ to each displayed mean. The relative columns are means of record-level ratios, so they are not obtained by taking ratios of the displayed marginal means.

The direct five-scale mean of the record-level RMS changes is -3.97% . Applying the square-root transformation only after averaging the five graph-dispersion effects gives $\sqrt{1 - 0.072834} - 1 = -3.71\%$. The two values differ because the square root and the averaging steps do not commute. In either form, a percentage change in squared pairwise differences is larger in magnitude than the corresponding change in RMS amplitude.

Table S3: Physical scale of graph dispersion in the evaluation summers. The RMS columns are $(2Q_h)^{1/2}$ transformations of the displayed mean Q_h columns. The interval is a 95% t -based interval that treats summers as independent. RMS change is the mean record-level relative change in D_h^{RMS} .

Bandwidth (km)	Q_h^M (°C ²)	Q_h^H (°C ²)	RMS^M (°C)	RMS^H (°C)	Q_h effect [interval] (%)	RMS change (%)
126	2.662	2.578	2.307	2.271	−2.86 [−5.04, −0.67]	−1.54
252	4.356	4.169	2.952	2.887	−3.94 [−6.50, −1.39]	−2.14
503	7.943	7.433	3.986	3.856	−5.91 [−8.91, −2.90]	−3.25
1,006	12.892	11.489	5.078	4.793	−10.44 [−13.49, −7.39]	−5.70
2,013	15.903	13.743	5.640	5.243	−13.27 [−16.28, −10.26]	−7.22

S5 Structural decompositions

S5.1 Climatology–anomaly energy

Let $\boldsymbol{\mu}_m$ be the site-specific 35-summer mean field for month m and write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$. For a fixed graph,

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \quad (\text{S9})$$

The first term is fixed within month m , so its high-minus-middle difference is exactly zero. Within each record, we divide the differences of the other two terms by the middle-day total Q_h mean. The resulting anomaly-energy and climatology–anomaly alignment components add to the original relative contrast before any aggregation.

Table S4: Exact climatology–anomaly decomposition. Entries are percentage points; brackets are 95% intervals. The climatology-energy difference is zero by construction.

Bandwidth (km)	Total effect [interval]	Anomaly energy [interval]	Climatology–anomaly alignment [interval]
126	−2.86 [−5.04, −0.67]	−2.63 [−3.55, −1.72]	−0.22 [−1.98, 1.54]
252	−3.94 [−6.50, −1.39]	−2.45 [−3.45, −1.45]	−1.49 [−3.69, 0.70]
503	−5.91 [−8.91, −2.90]	−1.42 [−2.43, −0.41]	−4.49 [−7.24, −1.74]
1,006	−10.44 [−13.49, −7.39]	−0.18 [−1.10, 0.74]	−10.26 [−13.20, −7.32]
2,013	−13.27 [−16.28, −10.26]	+0.41 [−0.50, 1.31]	−13.68 [−16.68, −10.67]

At 2,013 km, anomaly energy is lower in 16 of 33 summers, whereas the alignment component is negative in all 33. The corresponding counts at 1,006 km are 21 and 28. The broad-scale contraction is therefore associated with daily anomalies opposing the persistent monthly field, while the transient anomaly energy itself changes little. The decomposition is descriptive and conditional on the chosen climatology. The maximum aggregated identity error over the five graphs is 1.82×10^{-16} .

S5.2 Geographic and topographic bases

For a centred basis matrix $\mathbf{B}$, define the graph-orthogonal projection

$$\hat{\beta}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{L}_h \mathbf{y}_t, \quad \hat{\mathbf{y}}_{th} = \mathbf{B} \hat{\beta}_{th}.$$

The structured energy of $\hat{\mathbf{y}}_{th}$ and the residual energy of $\mathbf{y}_t - \hat{\mathbf{y}}_{th}$ add exactly to $Q_h(\mathbf{y}_t)$. Table S5 compares centred latitude, the latitude–longitude plane and a latitude–longitude–elevation basis. We obtained elevation from the invariant ERA5-Land geopotential field, selected the nearest 0.1-degree grid value at each analysis site and divided geopotential by 9.80665 m s^{-2} . The 121 elevations range from -0.9 to $1,940.3 \text{ m}$. Each basis column was centred and scaled before the projection. Such scaling leaves its column space unchanged.

Table S5: Nested spatial-basis sensitivity. Values are mean percentage-point contributions across the evaluation summers. The structured and residual columns for each basis add to the total, apart from rounding.

Bandwidth (km)	Total effect	Latitude structured	Latitude residual	Planar structured	Planar residual	Geographic– topographic structured	Geographic– topographic residual
126	−2.86	−2.58	−0.28	−2.81	−0.04	−1.16	−1.69
252	−3.94	−4.60	+0.66	−5.07	+1.13	−2.85	−1.10
503	−5.91	−8.38	+2.48	−9.32	+3.42	−6.13	+0.23
1,006	−10.44	−12.73	+2.29	−13.94	+3.50	−11.39	+0.94
2,013	−13.27	−14.96	+1.69	−16.23	+2.96	−14.34	+1.07

The latitude component is negative in 33 of 33 summers at every scale. The planar component is negative in 32 summers at 126 km and all 33 summers at the other four scales. At 2,013 km, its 95% interval is $[-18.16, -14.30]$ points and the residual interval is $[1.59, 4.33]$ points. Adding longitude makes the structured component more negative and the compensating residual more positive.

The geographic–topographic component is negative in 18, 21, 28, 33 and 33 of the 33 summers as bandwidth increases. At 126 km its estimate is -1.16 points with a 95% interval of $[-2.66, 0.33]$, which includes zero. At 2,013 km the estimate is -14.34 points $[-16.75, -11.92]$, and the residual is $+1.07$ points. The broad contraction persists after adding elevation to the latitude–longitude basis. The projection yields a combined geographic–topographic contribution; no order-invariant elevation contribution or physical mechanism follows from it. The maximum day-level identity error for the augmented basis is 8.88×10^{-14} , and the maximum aggregated error is 2.78×10^{-16} .

S6 Scale and support resolution

S6.1 Thirty-one-point bandwidth curve

The estimand is the equal-weight mean over the five prespecified bandwidths. We also evaluated 31 logarithmically spaced values between 125.800 and 2,012.796 km on the same

Table S6: The 31-point bandwidth curve. Values are mean graph-dispersion effects in per cent. The three column pairs continue the same ordered curve.

h (km)	Effect	h (km)	Effect	h (km)	Effect
125.8	-2.856	347.7	-4.596	876.1	-9.538
138.0	-2.958	381.4	-4.835	961.0	-10.151
151.3	-3.076	418.3	-5.125	1,054.0	-10.720
166.0	-3.211	458.8	-5.479	1,156.0	-11.239
182.1	-3.360	503.2	-5.907	1,268.0	-11.703
199.7	-3.521	551.9	-6.407	1,390.8	-12.113
219.0	-3.688	605.4	-6.974	1,525.4	-12.470
240.2	-3.859	664.0	-7.591	1,673.1	-12.778
263.5	-4.031	728.3	-8.239	1,835.1	-13.043
289.0	-4.206	798.8	-8.895	2,012.8	-13.268
317.0	-4.391				

121 sites, with the original event fields and labels. This exploratory profile describes the curve between the five bandwidths; only the prespecified values define the estimand.

All 30 adjacent changes are negative, and all 31 pointwise 95% intervals have upper endpoints below zero. The five prespecified estimates remain -2.86 , -3.94 , -5.91 , -10.44 and -13.27% . The curve shows no reversal at the 31 evaluated bandwidths. The intervals are descriptive pointwise summaries; no simultaneous confidence band is estimated.

S6.2 Four spatial-support configurations

The convergence comparison keeps the five physical bandwidths, primary peak times and original 121-site labels fixed. The 239-site configuration refines longitude while retaining the primary latitudes; the 236-site configuration refines latitude while retaining the primary longitudes. Each contains every primary site and is a subset of the 465-site grid, although the two one-direction refinements are not nested within one another.

Table S7: Four-configuration spatial-resolution comparison. Effects and the two error summaries are in percentage points. Differences use the 465-site result at the same fixed physical bandwidth as the reference.

Configuration	Sites	126	252	503	1,006	2,013	Max. $ \Delta $	RMSE
Primary support	121	-2.856	-3.945	-5.907	-10.441	-13.268	0.991	0.764
Longitude refined	239	-2.839	-3.846	-5.968	-10.552	-13.351	0.929	0.728
Latitude refined	236	-3.129	-4.308	-6.529	-10.865	-13.458	0.500	0.384
Dense support	465	-3.395	-4.543	-6.898	-11.340	-13.959	0.000	0.000

Every support retains the broad-to-local ordering and a negative effect at all five bandwidths. Relative to 465 sites, the 121-site profile has a maximum absolute discrepancy of 0.99 points and an across-scale RMSE of 0.76 points. These finite-grid comparisons quantify sensitivity to the available spatial support. Their scope is finite-support sensitivity, with continuous-domain convergence left unassessed. Intersite spacing and the separate station comparison provide the least support for the narrowest graph.

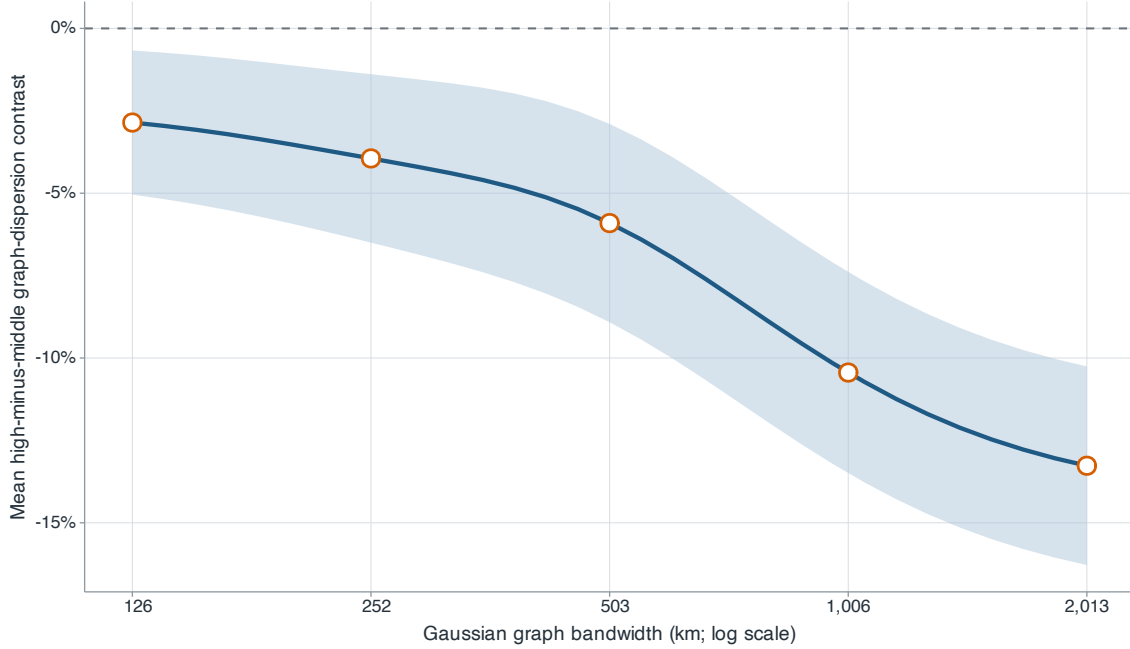


Figure 1: Bandwidth-resolution profile on the 121-site support. The blue line and shaded band show the 31-point profile and its pointwise 95% intervals; orange open circles mark the five prespecified bandwidths.

Alt text: A line descends smoothly as bandwidth increases from 126 to 2,013 km. All 31 estimated effects are negative, and the five prespecified bandwidths lie on the same monotone profile.

S7 Historical extension, 1950–1990

The ERA5-Land archive distributed through the Copernicus Climate Data Store begins in January 1950.¹ The historical-extension plan was finalised on 9 August 2026, after the 1991–2025 analysis and before retrieval of any 1950–1990 value. It specified all 41 JJA summers, the same 121 sites, the five bandwidths and the original quality control, Bolton wet-bulb calculation, UTC daily-peak rule and type-7 within-record quartiles. The analysis weights the three months, five scales and summers equally. The extension is separated in calendar time from the 1991–2025 period, but both analyses share the ERA5-Land model and assimilation system. The dated plan also specified one 99,999-draw joint product cyclic-shift calculation with seed 20260810, with all five dispersion columns rotated together within each historical month-year record.

The five-scale effect over 1950–1990 is -11.04% (95% interval $[-12.84, -9.24]\%$), with a negative estimate in 40 of 41 summers. Deleting one summer at a time gives estimates from -11.33% to -10.62% . Table S8 gives the fixed period split and the five-scale profile.

Provider documentation identifies the forcing change used for the period split: ERA5-Land for 1950–1978 used the preliminary ERA5 back extension, whereas the later period used the consolidated ERA5 release.² Both period estimates are negative, and their intervals overlap.

¹Dataset overview: <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview>.

²ERA5-Land production note: <https://confluence.ecmwf.int/pages/viewpage.action?pageId=40>

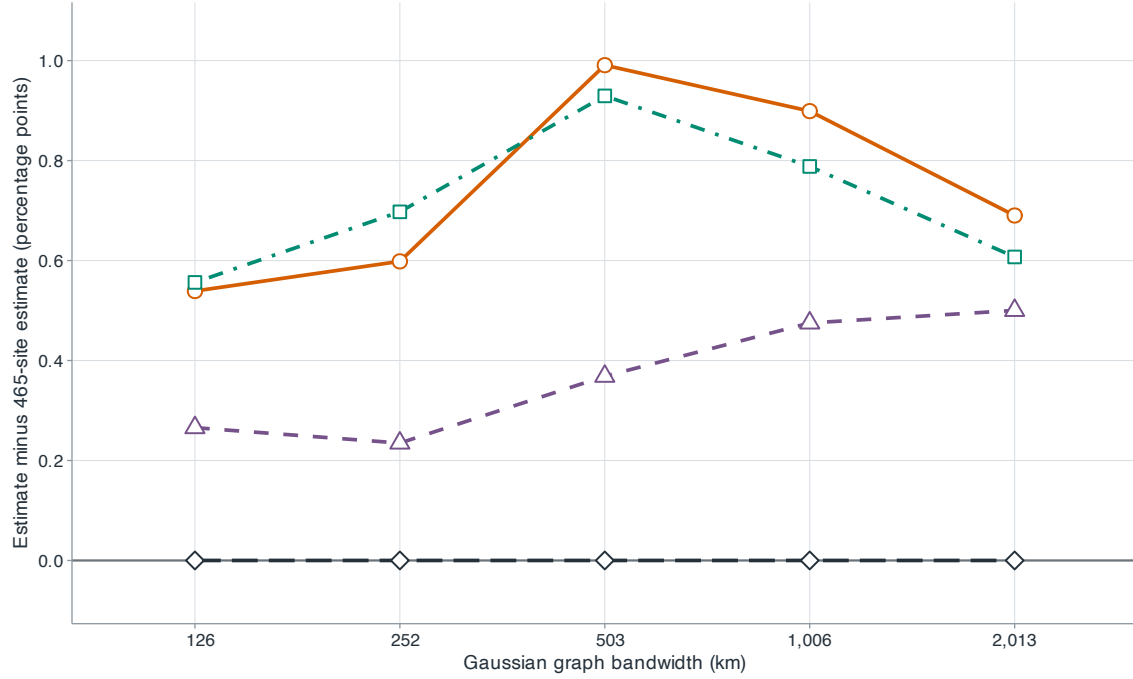


Figure 2: Differences from the 465-site estimate at five fixed bandwidths. Orange circles with a solid line denote the primary 121-site support; purple triangles with a dashed line, the latitude-refined 236-site support; teal squares with a dot-dash line, the longitude-refined 239-site support; and dark diamonds with a long-dash line, the dense 465-site reference.

Alt text: Four lines compare each spatial support with the 465-site reference across five bandwidths. All differences are between zero and one percentage point, and the broad-to-local ordering of the underlying effects is unchanged.

Table S8: The 1950–1990 historical extension. Estimates and intervals are percentages; the final column counts summers with a negative estimate. Period rows average the five bandwidths. Scale rows use all 41 summers.

Period or bandwidth	Summers	Estimate	95% interval	Negative summers
1950–1990 overall	41	−11.04	[−12.84, −9.24]	40
1950–1978	29	−10.50	[−12.50, −8.51]	28
1979–1990	12	−12.34	[−16.65, −8.04]	12
126 km	41	−5.96	[−7.37, −4.54]	38
252 km	41	−7.51	[−9.15, −5.87]	39
503 km	41	−10.08	[−12.11, −8.04]	40
1,006 km	41	−14.50	[−16.72, −12.27]	41
2,013 km	41	−17.17	[−19.44, −14.91]	41

The 1950–1978 estimate inherits uncertainty from the preliminary forcing data.

As specified in the pre-access extension plan, we applied the product cyclic-shift calculation to 123 month-year records, using $B = 99,999$ joint five-scale shifts and seed 20260810. None of the sampled statistics was at or below the observed -11.043% , so the plus-one lower-tail value is 0.00001. The sampled null mean and standard deviation are 0.853% and 1.436 percentage points. Every scale has an unadjusted value of 0.00001; the complete-null single-step values range from 0.00001 to 0.00006.

For the exploratory historical climatology–anomaly decomposition, the monthly reference field was estimated within 1950–1990:

$$\mu_m^H = \frac{1}{41} \sum_{y=1950}^{1990} \left(\frac{1}{n_m} \sum_{t \in m} y_{yt} \right), \quad \mathbf{a}_{yt}^H = y_{yt} - \mu_m^H.$$

Thus μ_m^H is the site-specific mean field for month m across the 41 historical summers. The anomaly and alignment terms therefore describe departures from the historical period’s own persistent monthly spatial pattern.

Table S9 reports the protocol-specified latitude and planar accounting and the exploratory climatology–anomaly accounting for the historical extension. The latitude and planar structured components are negative in all 41 summers at every bandwidth. At 2,013 km, anomaly energy is -0.28 points with interval $[-1.01, 0.45]$, while the climatology–anomaly alignment component is -16.89 points $[-19.31, -14.47]$ and is negative in all 41 summers. Thus the broad-scale historical result has a low-dimensional and alignment-dominated accounting pattern similar to that in the 1991–2025 period.

Table S9: Structural accounting for the 1950–1990 extension. Entries are percentage points. Within each geographic basis, the structured and residual columns add to the total. The anomaly and alignment columns also add to the total because the fixed climatology–energy difference is zero.

Bandwidth (km)	Total effect	Latitude structured	Latitude residual	Planar structured	Planar residual	Anomaly energy	Climatology– anomaly alignment
126	−5.96	−2.83	−3.13	−3.15	−2.81	−3.22	−2.74
252	−7.51	−4.92	−2.59	−5.50	−2.01	−3.30	−4.21
503	−10.08	−8.86	−1.21	−9.92	−0.15	−2.41	−7.67
1,006	−14.50	−13.60	−0.90	−14.99	+0.49	−1.02	−13.48
2,013	−17.17	−16.13	−1.05	−17.62	+0.44	−0.28	−16.89

S8 NOAA station extension in ten evaluation years

The station-extension plan was finalised on 9 August 2026 before retrieval of the new station-year files or calculation of their high–middle effects. The plan fixed the years by enumeration rather than by a rule applied to station outcomes: 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025. The set can be written as $\{1992 + 4k : k = 0, \dots, 7\} \cup \{2023, 2025\}$. The protocol defines the year set by this enumeration; station coverage and effect estimates

do not enter the choice. The 2015 and 2022 development years are excluded. We archived the official Integrated Surface Database (ISD) station-history snapshot used for selection.³ The planned eligibility rule required a station-history end date on or after 31 August 2025, but that snapshot contained no such candidate: its global maximum end date was 28 August and its maximum inside the fixed study rectangle was 24 August. The operational administrative cutoff was therefore 24 August. This outcome-blind metadata amendment was made before retrieval of station WBT observations, matched ERA5-Land values or calculation of any graph contrast. The year set, study rectangle, geographic proximity rule and target sample size were unchanged.

For a candidate station s , the proximity criterion was

$$d(s, \mathcal{S}_{121}) = \min_{i \in \mathcal{S}_{121}} d(s, i) \leq 150 \text{ km},$$

where $\mathcal{S}_{121}$ is the fixed primary analysis support. For coordinate differences in degrees, the pairwise distance was

$$d(s, i) = \left[\{111.32 \cos(\bar{\phi}_{si})(\lambda_s - \lambda_i)\}^2 + \{110.57(\phi_s - \phi_i)\}^2 \right]^{1/2},$$

with $\bar{\phi}_{si}$, the pair’s mean latitude, converted to radians in the cosine. This filter was applied before maximin selection and kept station locations close to the primary spatial support. Of the 228 stations satisfying the rectangle and date conditions, 20 exceeded 150 km, leaving 208 candidates; the largest retained nearest-site distance was 147.73 km and the smallest excluded distance was 155.72 km. A deterministic maximin rule based on station coordinates selected 30 stations. The study rectangle defined the candidate domain. A station-year qualified with at least 400 exact UTC hours having temperature, dew point and usable station pressure; 26–30 of the 30 selected stations qualified in each year. We calculated station wet-bulb temperature with the same Bolton implementation used for ERA5-Land.

The nearest ERA5-Land cells for the Shengsi and Taidong island stations had no finite land values at any requested time. We recorded both point series as unavailable. The other 28 station locations had finite ERA5-Land series. Combining this land-mask restriction with the hourly qualification rule left 24–28 matched stations per year and 175,172 paired hours. Table S10 reports the measurement comparison. Bias is ERA5-Land minus station wet-bulb temperature; the centred correlation subtracts each station mean before pooling. The all-year row pools all matched hours across years.

For the effect comparison, ERA5-Land peak times and high/middle labels defined the event days, and we aligned station observations to those events. Each graph field used the same available station subset for NOAA and ERA5-Land and required at least ten stations. All 30 month-year records retained at least one high and one middle field, giving 345 usable event fields. We formed a high-to-middle ratio within record, averaged the three months equally within year and then averaged the ten years. Table S11 reports 95% intervals. Inference focuses on the 503-, 1,006- and 2,013-km graphs supported by the station spacing; the two narrower graphs remain descriptive.

³NOAA Integrated Surface Database: <https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>.

Table S10: ERA5-Land and NOAA station wet-bulb agreement in ten evaluation years. Bias, MAE and RMSE are in °C. Stations are the qualified locations with a finite nearest-cell ERA5-Land series in that year.

Year	Stations	Matched hours	Bias	MAE	RMSE	Centred correlation
1992	25	17,887	-0.25	0.88	1.16	0.919
1996	25	19,147	-0.33	0.88	1.16	0.908
2000	25	16,765	-0.40	0.94	1.21	0.908
2004	25	16,829	-0.20	0.91	1.18	0.923
2008	25	17,310	-0.20	0.88	1.14	0.909
2012	25	17,349	-0.27	0.93	1.22	0.918
2016	24	16,589	-0.34	0.88	1.13	0.937
2020	28	19,323	-0.26	1.04	1.48	0.897
2023	27	16,554	-0.19	1.02	1.49	0.925
2025	28	17,419	-0.17	1.01	1.45	0.933
All years	28	175,172	-0.26	0.94	1.27	0.916

Table S11: High-middle graph-dispersion effects at ERA5-Land-defined peak times and labels in ten evaluation years. Effects and intervals are percentages. “Negative” counts years with a negative scale-specific estimate. Each scale contains all 30 month-year records and an average of 25.3 stations per event field.

Bandwidth (km)	NOAA effect [95% interval]	NOAA negative years	ERA5-Land effect [95% interval]	ERA negative years
126	-6.23 [-25.87, 13.41]	6/10	-11.60 [-27.49, 4.29]	7/10
252	-7.23 [-21.01, 6.55]	7/10	-12.74 [-22.68, -2.81]	8/10
503	-12.50 [-24.05, -0.96]	8/10	-17.13 [-25.15, -9.11]	10/10
1,006	-18.03 [-29.17, -6.90]	9/10	-21.60 [-29.07, -14.14]	10/10
2,013	-21.47 [-32.51, -10.43]	9/10	-24.23 [-31.72, -16.74]	10/10

The station and paired ERA5-Land profiles are negative at all five scales and become more negative with bandwidth. Averaging the three station-supported scales within each year gives -17.34% (95% interval [-28.49, -6.18]%), with a negative average in nine of ten years. The same fixed broad-profile calculation for the matched ERA5-Land fields gives -20.99% ([-28.57, -13.40]%), negative in all ten years. These results compare the two fields on identical, time-varying station supports. Because ERA5-Land defines the event times and labels, this is an external measurement and temporally separated effect comparison under a common event definition, not an independent event replication. The irregularly sampled years, changing qualified subsets and sparse support at 126 and 252 km limit the comparison.

S9 Continuous intensity and calendar matching

We evaluated two alternative event contrasts using the same sites, graphs and aggregation hierarchy. For the continuous calculation, we regressed $\log Q_h$ on centred regional wet-bulb temperature separately in each month-year record. We then averaged slopes equally across months, scales and the 33 evaluation summers. The five-scale mean slope is -0.0650 °C⁻¹, with a 95% interval of [-0.0822, -0.0479] and negative summer slopes in 31 of 33 years.

The calendar-matched calculation compares each high day with the mean of middle days no more than three or five calendar days away in the same month-year record. We average eligible high-day contrasts within record before applying the original equal-month, equal-scale and equal-summer hierarchy. The ± 3 -day window matches 670 of 792 high days (84.6%); the ± 5 -day window matches 750 (94.7%). Each of the 99 evaluation month-year records contributes at least one match under both windows.

Table S12: Continuous-intensity and calendar-matched results for the 33 evaluation summers. Continuous entries are slopes of $\log Q_h$ per $^{\circ}\text{C}$. Matching entries are percentage effects. Brackets give 95% intervals.

Bandwidth	Continuous slope [interval]	± 3 -day match [interval]	± 5 -day match [interval]
126	-0.0318 [-0.0435, -0.0201]	-2.32 [-4.73, 0.09]	-2.54 [-4.90, -0.18]
252	-0.0408 [-0.0553, -0.0263]	-3.29 [-6.03, -0.56]	-3.54 [-6.18, -0.90]
503	-0.0546 [-0.0735, -0.0356]	-4.41 [-7.44, -1.38]	-4.74 [-7.66, -1.83]
1,006	-0.0877 [-0.1092, -0.0661]	-7.17 [-10.32, -4.02]	-7.90 [-10.94, -4.86]
2,013	-0.1104 [-0.1325, -0.0882]	-8.87 [-12.06, -5.68]	-9.89 [-12.98, -6.80]
Five-scale mean	-0.0650 [-0.0822, -0.0479]	-5.21 [-8.01, -2.41]	-5.72 [-8.43, -3.02]

The matched effects are weaker than the unrestricted high-to-middle contrast, but their broad-to-local ordering remains. Calendar matching also changes the target population of day pairs: only high days with eligible middle days enter the high-day average.

S10 Heavy-tailed ratio stress test

We compared the raw effect $r - 1$, the log effect $\log r$ and the bounded symmetric effect $2(r - 1)/(r + 1)$, where r is the high-to-middle dispersion ratio. The stress design uses the 121 observed site locations, exponential spatial covariance with 450-km range and 5% nugget, circular field correlation 0.45 and regional-mean autoregression 0.65. A common daily scale $\sqrt{3/\chi_3^2}$ produces heavy-tailed graph energy. Each scenario has 2,000 paired replications of six 30–31-day records. All three measures use the same 999 product shifts within a replication; the seed is 20260811.

Table S13: Raw, log and bounded effect measures under the heavy-tailed stress design. Bias and RMSE are on each measure’s scale. Size and power are lower-tail rejection proportions at 0.05. Alternative targets correspond to a high-day energy multiplier of 0.93.

Measure	Null bias	Null RMSE	Size	Alternative target	Alt. bias	Alt. RMSE	Power
Raw ratio	0.3716	1.4725	0.0470	-0.0700	0.3456	1.3694	0.0795
Log ratio	-0.0698	0.3283	0.0490	-0.0726	-0.0698	0.3283	0.0850
Bounded symmetric	-0.0657	0.2771	0.0475	-0.0725	-0.0574	0.2748	0.0790

The three product-shift tests retain near-nominal size in this design. The raw ratio nevertheless has an RMSE more than five times that of the bounded measure under the null. Randomisation validity governs label assignment for a fixed statistic, whereas ratio-estimation stability also depends on the right tail of daily energy. The log and bounded summaries reduce that instability, with the bounded measure giving the smallest RMSE in these cells.

S11 Simulation specifications and results

S11.1 Coupled label–field stress designs

The coupled simulation uses the 121 observed site coordinates and six records of lengths 30, 31, 30, 30, 31 and 30 days. Isotropic anomaly fields have exponential covariance with 450-km range and 5% nugget. The field state has autoregressive coefficient 0.45, and the regional weather series has coefficient 0.65. High and middle labels use type-7 quartiles within each record. A standardised spatial gradient is proportional to $0.8x + y$, where x and y are projected site coordinates. Its sign is driven by $0.75\tilde{\mu}_t + (1 - 0.75^2)^{1/2}Z_t$, so labels and spatial structure share a latent weather component even under the null.

Table S14: Exact changes defining the eight coupled label–field designs. Every cell retains the common six-record construction and within-record labels.

Design	Null	Contraction alternative
Shared latent gradient	Gradient amplitude is 1.8 on every day; its sign shares the latent driver with regional weather.	Gradient amplitude on high days is reduced from 1.8 to 1.2.
Within-month progression	Regional mean is a linear trend from -1.4 to 1.4 plus 0.60 times the AR(1) weather series; gradient amplitude remains 1.8.	Gradient amplitude declines linearly from 1.95 to 1.30 across the record.
Anisotropic covariance	Anomaly e-folding ranges are 800 km in projected longitude and 300 km in projected latitude.	High-day anomaly amplitude is multiplied by 0.94.
Peak-hour selection	Each daily regional mean is the maximum of 24 hourly values with sinusoidal amplitude 1.8 and independent noise SD 0.35; anomaly covariance is hour-invariant.	High-day anomaly amplitude is multiplied by 0.94.

Each field is evaluated with the five Gaussian graphs and a five-bin variogram profile whose bins are the quintiles of the 7,260 observed pair distances. For each replication, the target is the high-to-middle contrast in the conditional expected quadratic forms. The graph and variogram statistics use the same simulated fields, labels and product shifts. Each of the eight cells contains 1,000 replications with 499 shifts per test.

S11.2 Repeated-summer experiment

The annual experiment crosses $R \in \{20, 33, 60, 120\}$, $\rho \in \{0, 0.3, 0.6\}$, Gaussian, centred log-normal and standardised t_3 innovations, and effects $\delta \in \{0, -0.035, -0.07\}$. The resulting 108 cells each contain 10,000 replications after a 200-step burn-in. For retained summer r ,

$$U_r = \rho U_{r-1} + (1 - \rho^2)^{1/2} \varepsilon_r, \quad Y_r = \delta + 0.12 U_r.$$

The Gaussian innovation is standard normal. For $Z \sim N(0, 1)$, the centred unit-variance log-normal innovation is

$$\frac{\exp(0.65Z) - \exp(0.65^2/2)}{\{[\exp(0.65^2) - 1] \exp(0.65^2)\}^{1/2}},$$

and the heavy-tailed innovation is $T_3/\sqrt{3}$. The point estimate is $\bar{Y}$. The ordinary interval uses the sample variance and a t_{R-1} critical value. The lag-2 calculation uses a Bartlett long-run variance with the same critical value. The growing-lag interval uses $L = \max\{1, \lfloor R^{1/3} \rfloor\}$ and a standard-normal critical value. The sign test uses the exact binomial upper tail of the number of negative summers. The three-test decision requires the Student, lag-2 and sign p -values all to be at most 0.025. Standardised diagnostics use the known long-run variance $0.12^2(1 + \rho)/(1 - \rho)$.

The following tables report all eight coupled cells and all 108 annual cells.

Table S15: Complete results for the eight coupled label-field stress designs. Targets, bias and RMSE are percentage points; rejection is per cent over 1,000 replications. Graph and five-bin variogram profiles use identical fields, labels and product shifts.

Scenario	Family	Graph profile				Five-bin variogram			
		Target	Bias	RMSE	Reject	Target	Bias	RMSE	Reject
Shared-latent null	Null	0.00	0.38	3.63	5.5	-0.00	0.53	4.58	5.3
Shared-latent contraction	Alternative	-27.37	0.19	2.70	100.0	-36.66	0.25	3.26	100.0
Seasonal-progression null	Null	0.00	0.64	3.76	4.0	-0.00	0.85	4.77	4.3
Seasonal-gradient contraction	Alternative	-10.74	0.27	3.65	78.6	-14.72	0.33	4.60	83.3
Anisotropic null	Null	0.00	0.31	3.86	7.0	0.00	0.42	4.81	7.3
Anisotropic amplitude contraction	Alternative	-5.92	0.21	3.63	51.4	-3.98	0.32	4.58	25.5
Peak-selection null	Null	0.00	0.21	3.69	6.6	-0.00	0.34	4.63	6.6
Peak-selection contraction	Alternative	-5.90	0.36	3.50	54.2	-3.96	0.52	4.46	25.1

Table S16: Complete 108-cell repeated-summer results: estimation, coverage and standardised-mean diagnostics. Bias and RMSE are percentage points; coverage is per cent. L is the growing Bartlett lag. The standardised columns use the known long-run variance.

R	ρ	Innovation	Effect	Bias percentage points	RMSE percentage points	L	Student	Lag 2 coverage (%)	Growing	z mean	z SD	$z_{.025}, z_{.975}$
20	0.0	Gaussian	Null	0.015	2.655	2	95.4	91.7	89.9	0.006	0.990	-1.94, 1.98
20	0.0	Gaussian	-3.5%	-0.015	2.680	2	94.8	91.1	89.2	-0.006	0.999	-1.98, 1.98
20	0.0	Gaussian	-7%	-0.015	2.665	2	95.1	91.3	89.4	-0.006	0.993	-1.92, 1.99
20	0.0	Skewed	Null	-0.046	2.691	2	91.2	87.6	85.9	-0.017	1.003	-1.71, 2.21
20	0.0	Skewed	-3.5%	0.022	2.676	2	92.1	88.7	86.9	0.008	0.997	-1.71, 2.18
20	0.0	Skewed	-7%	-0.024	2.662	2	91.8	88.3	86.5	-0.009	0.992	-1.68, 2.14
20	0.0	t_3	Null	-0.028	2.672	2	95.3	91.7	89.7	-0.010	0.996	-1.92, 1.95
20	0.0	t_3	-3.5%	0.028	2.599	2	95.9	92.0	90.4	0.011	0.968	-1.90, 1.90
20	0.0	t_3	-7%	0.028	2.696	2	95.9	92.0	90.1	0.010	1.005	-1.88, 1.91
20	0.3	Gaussian	Null	0.081	3.584	2	85.5	86.8	84.8	0.022	0.980	-1.91, 1.94
20	0.3	Gaussian	-3.5%	-0.096	3.597	2	85.0	86.6	84.2	-0.026	0.983	-1.96, 1.93
20	0.3	Gaussian	-7%	0.036	3.620	2	85.0	86.5	84.0	0.010	0.990	-1.93, 1.92
20	0.3	Skewed	Null	0.053	3.612	2	82.7	84.4	82.3	0.014	0.988	-1.66, 2.18
20	0.3	Skewed	-3.5%	0.019	3.615	2	82.6	84.2	82.2	0.005	0.989	-1.68, 2.17
20	0.3	Skewed	-7%	-0.038	3.545	2	83.0	84.6	82.3	-0.010	0.969	-1.68, 2.12
20	0.3	t_3	Null	0.056	3.486	2	85.9	87.5	85.1	0.015	0.953	-1.85, 1.91
20	0.3	t_3	-3.5%	-0.126	3.533	2	85.7	87.3	84.8	-0.035	0.966	-1.94, 1.85
20	0.3	t_3	-7%	-0.002	3.620	2	85.8	87.8	85.3	-0.001	0.990	-1.92, 1.91
20	0.6	Gaussian	Null	-0.042	5.096	2	66.5	76.2	73.4	-0.008	0.950	-1.89, 1.83
20	0.6	Gaussian	-3.5%	0.002	5.128	2	67.0	76.3	73.8	0.000	0.956	-1.89, 1.85
20	0.6	Gaussian	-7%	0.000	5.175	2	66.6	75.9	73.2	0.000	0.964	-1.91, 1.88
20	0.6	Skewed	Null	0.038	5.187	2	64.4	73.9	71.2	0.007	0.967	-1.64, 2.10
20	0.6	Skewed	-3.5%	-0.043	5.140	2	64.5	74.3	71.5	-0.008	0.958	-1.62, 2.11
20	0.6	Skewed	-7%	-0.039	5.115	2	65.1	74.6	71.7	-0.007	0.953	-1.60, 2.11
20	0.6	t_3	Null	0.047	4.987	2	65.9	76.3	73.5	0.009	0.929	-1.77, 1.81
20	0.6	t_3	-3.5%	0.075	5.101	2	65.1	75.8	72.7	0.014	0.950	-1.81, 1.87
20	0.6	t_3	-7%	-0.026	5.254	2	65.5	76.7	73.7	-0.005	0.979	-1.84, 1.79
33	0.0	Gaussian	Null	0.057	2.065	3	95.2	93.3	91.2	0.027	0.988	-1.92, 1.97
33	0.0	Gaussian	-3.5%	-0.004	2.078	3	95.2	93.1	91.2	-0.002	0.995	-1.95, 1.92
33	0.0	Gaussian	-7%	-0.021	2.062	3	95.2	93.5	91.4	-0.010	0.987	-1.93, 1.93
33	0.0	Skewed	Null	-0.029	2.071	3	92.4	90.6	88.5	-0.014	0.991	-1.75, 2.12
33	0.0	Skewed	-3.5%	-0.002	2.061	3	92.9	91.0	89.0	-0.001	0.987	-1.72, 2.11
33	0.0	Skewed	-7%	0.005	2.080	3	92.6	90.7	88.8	0.002	0.996	-1.73, 2.13
33	0.0	t_3	Null	-0.019	2.089	3	95.8	93.7	91.5	-0.009	1.000	-1.93, 1.92
33	0.0	t_3	-3.5%	0.007	2.062	3	95.6	93.2	91.0	0.003	0.987	-1.93, 1.93
33	0.0	t_3	-7%	-0.004	2.127	3	95.7	93.6	91.5	-0.002	1.018	-1.94, 1.91
33	0.3	Gaussian	Null	-0.046	2.837	3	85.4	89.0	87.3	-0.016	0.996	-1.99, 1.95
33	0.3	Gaussian	-3.5%	-0.002	2.819	3	85.4	88.6	87.2	-0.001	0.990	-1.93, 1.94
33	0.3	Gaussian	-7%	0.018	2.808	3	85.7	88.8	87.3	0.006	0.987	-1.92, 1.92
33	0.3	Skewed	Null	0.000	2.795	3	83.3	87.2	85.9	0.000	0.982	-1.72, 2.11
33	0.3	Skewed	-3.5%	-0.004	2.834	3	82.9	86.4	85.1	-0.001	0.995	-1.71, 2.13
33	0.3	Skewed	-7%	-0.008	2.792	3	83.4	86.8	85.4	-0.003	0.981	-1.73, 2.09

Continued on next page

Table S16 continued

R	ρ	Innovation	Effect	Bias	RMSE	L	Student	Lag 2	Growing	z mean	z SD	$z_{.025}, z_{.975}$
33	0.3	t_3	Null	-0.011	2.773	3	85.0	89.2	87.5	-0.004	0.974	-1.93, 1.89
33	0.3	t_3	-3.5%	0.012	2.804	3	85.5	89.4	88.2	0.004	0.985	-1.92, 1.94
33	0.3	t_3	-7%	0.033	2.760	3	85.4	89.1	87.8	0.012	0.970	-1.94, 1.91
33	0.6	Gaussian	Null	0.004	4.112	3	66.1	78.7	79.0	0.001	0.984	-1.92, 1.92
33	0.6	Gaussian	-3.5%	0.031	4.059	3	66.8	79.1	79.2	0.008	0.972	-1.89, 1.87
33	0.6	Gaussian	-7%	0.027	4.049	3	67.3	79.5	79.6	0.006	0.969	-1.90, 1.88
33	0.6	Skewed	Null	0.020	4.090	3	64.9	77.6	77.9	0.005	0.979	-1.70, 2.12
33	0.6	Skewed	-3.5%	0.043	4.053	3	65.6	77.8	78.0	0.010	0.970	-1.68, 2.10
33	0.6	Skewed	-7%	0.037	4.070	3	65.0	77.7	77.7	0.009	0.974	-1.68, 2.09
33	0.6	t_3	Null	-0.014	3.994	3	66.5	79.8	80.1	-0.003	0.956	-1.87, 1.83
33	0.6	t_3	-3.5%	-0.054	4.004	3	66.0	79.9	80.3	-0.013	0.958	-1.89, 1.82
33	0.6	t_3	-7%	0.006	4.158	3	65.4	79.6	79.8	0.001	0.995	-1.96, 1.88
60	0.0	Gaussian	Null	0.002	1.554	3	95.1	94.0	92.9	0.001	1.003	-1.93, 1.95
60	0.0	Gaussian	-3.5%	-0.018	1.562	3	94.8	93.6	92.4	-0.012	1.008	-2.01, 1.94
60	0.0	Gaussian	-7%	-0.003	1.551	3	95.0	93.8	92.8	-0.002	1.001	-1.94, 1.98
60	0.0	Skewed	Null	-0.002	1.553	3	93.4	92.2	91.1	-0.001	1.003	-1.79, 2.10
60	0.0	Skewed	-3.5%	0.020	1.541	3	93.5	92.5	91.4	0.013	0.995	-1.79, 2.09
60	0.0	Skewed	-7%	0.039	1.545	3	94.1	93.2	92.0	0.025	0.997	-1.78, 2.15
60	0.0	t_3	Null	0.019	1.583	3	95.4	94.3	93.2	0.013	1.022	-1.96, 2.01
60	0.0	t_3	-3.5%	0.008	1.523	3	95.5	94.3	93.3	0.005	0.983	-1.92, 1.96
60	0.0	t_3	-7%	-0.008	1.536	3	95.7	94.8	93.6	-0.005	0.991	-1.89, 1.92
60	0.3	Gaussian	Null	0.016	2.097	3	85.0	90.0	89.7	0.008	0.993	-1.99, 1.92
60	0.3	Gaussian	-3.5%	-0.016	2.068	3	85.9	90.9	90.4	-0.007	0.979	-1.92, 1.90
60	0.3	Gaussian	-7%	0.033	2.083	3	85.6	90.6	90.3	0.015	0.986	-1.91, 1.91
60	0.3	Skewed	Null	-0.019	2.085	3	84.2	89.0	88.8	-0.009	0.988	-1.81, 2.07
60	0.3	Skewed	-3.5%	0.018	2.096	3	84.1	89.1	88.8	0.008	0.993	-1.78, 2.13
60	0.3	Skewed	-7%	-0.001	2.097	3	83.7	88.6	88.4	-0.001	0.993	-1.83, 2.08
60	0.3	t_3	Null	-0.014	2.082	3	85.2	90.5	90.0	-0.007	0.986	-1.92, 1.98
60	0.3	t_3	-3.5%	0.035	2.088	3	84.7	90.7	90.3	0.017	0.989	-1.91, 1.96
60	0.3	t_3	-7%	-0.010	2.087	3	85.1	90.5	90.1	-0.005	0.988	-1.96, 1.95
60	0.6	Gaussian	Null	-0.011	3.056	3	67.0	81.0	82.5	-0.003	0.986	-1.92, 1.94
60	0.6	Gaussian	-3.5%	-0.029	3.066	3	66.9	81.5	83.0	-0.009	0.989	-1.96, 1.90
60	0.6	Gaussian	-7%	0.030	3.036	3	67.3	81.3	82.8	0.010	0.980	-1.89, 1.92
60	0.6	Skewed	Null	0.009	3.026	3	65.9	80.1	81.5	0.003	0.977	-1.78, 2.06
60	0.6	Skewed	-3.5%	-0.016	3.062	3	65.4	79.9	81.5	-0.005	0.988	-1.77, 2.10
60	0.6	Skewed	-7%	-0.018	3.038	3	66.5	80.2	81.6	-0.006	0.980	-1.76, 2.05
60	0.6	t_3	Null	-0.003	2.988	3	66.1	81.0	82.6	-0.001	0.964	-1.90, 1.87
60	0.6	t_3	-3.5%	0.005	3.063	3	66.6	81.3	82.9	0.002	0.989	-1.89, 1.92
60	0.6	t_3	-7%	0.026	3.046	3	66.0	81.5	83.0	0.008	0.983	-1.94, 1.92
120	0.0	Gaussian	Null	0.013	1.098	4	94.8	94.3	93.7	0.012	1.003	-1.98, 1.97
120	0.0	Gaussian	-3.5%	0.003	1.100	4	94.9	94.4	93.7	0.002	1.004	-1.93, 1.99
120	0.0	Gaussian	-7%	0.023	1.101	4	94.7	94.2	93.2	0.021	1.005	-1.94, 2.02
120	0.0	Skewed	Null	0.011	1.091	4	94.4	93.7	93.0	0.010	0.996	-1.82, 2.07
120	0.0	Skewed	-3.5%	0.003	1.084	4	94.2	93.5	93.1	0.003	0.989	-1.86, 2.05
120	0.0	Skewed	-7%	-0.004	1.101	4	93.7	93.3	92.5	-0.003	1.005	-1.88, 2.07

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Table S16 continued

R	ρ	Innovation	Effect	Bias	RMSE	L	Student	Lag 2	Growing	z mean	z SD	$z_{.025}, z_{.975}$
120	0.0	t_3	Null	0.008	1.115	4	95.2	94.5	93.9	0.007	1.018	-1.95, 2.01
120	0.0	t_3	-3.5%	-0.007	1.099	4	95.2	94.4	93.8	-0.006	1.003	-1.96, 1.96
120	0.0	t_3	-7%	0.004	1.100	4	95.5	94.8	94.0	0.004	1.004	-1.92, 2.02
120	0.3	Gaussian	Null	0.013	1.493	4	84.8	90.8	91.5	0.008	1.000	-1.98, 1.95
120	0.3	Gaussian	-3.5%	0.001	1.471	4	85.6	91.2	91.8	0.001	0.986	-1.92, 1.94
120	0.3	Gaussian	-7%	0.002	1.489	4	84.8	90.9	91.5	0.001	0.997	-1.94, 1.95
120	0.3	Skewed	Null	-0.004	1.502	4	83.6	89.6	90.2	-0.003	1.006	-1.86, 2.10
120	0.3	Skewed	-3.5%	-0.014	1.485	4	84.7	90.2	90.7	-0.009	0.995	-1.87, 2.06
120	0.3	Skewed	-7%	0.025	1.491	4	84.2	90.0	90.4	0.017	0.999	-1.82, 2.06
120	0.3	t_3	Null	-0.009	1.495	4	84.7	91.1	91.7	-0.006	1.001	-1.94, 1.93
120	0.3	t_3	-3.5%	-0.028	1.475	4	84.6	91.2	91.7	-0.018	0.988	-1.95, 1.94
120	0.3	t_3	-7%	-0.002	1.488	4	84.5	90.7	91.4	-0.001	0.997	-1.98, 1.94
120	0.6	Gaussian	Null	-0.018	2.164	4	67.9	82.8	86.5	-0.008	0.988	-1.97, 1.95
120	0.6	Gaussian	-3.5%	-0.010	2.171	4	67.8	82.7	86.6	-0.005	0.991	-1.97, 1.94
120	0.6	Gaussian	-7%	0.023	2.169	4	66.9	82.3	86.5	0.010	0.990	-1.92, 1.93
120	0.6	Skewed	Null	-0.012	2.162	4	67.2	81.8	85.6	-0.005	0.987	-1.83, 2.09
120	0.6	Skewed	-3.5%	-0.020	2.181	4	65.7	81.3	85.6	-0.009	0.996	-1.85, 2.06
120	0.6	Skewed	-7%	0.014	2.180	4	66.5	81.6	85.5	0.006	0.995	-1.82, 2.08
120	0.6	t_3	Null	-0.013	2.184	4	66.6	82.5	86.4	-0.006	0.997	-1.87, 1.91
120	0.6	t_3	-3.5%	-0.053	2.146	4	66.5	82.5	86.6	-0.024	0.979	-1.97, 1.90
120	0.6	t_3	-7%	0.015	2.180	4	65.7	82.2	86.3	0.007	0.995	-1.95, 1.93

Table S17: Complete 108-cell repeated-summer results: one-sided rejection percentages. The three-test column requires Student, lag-2 and sign values all to be at most 0.025. Under non-null cells the entries are power, not size.

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
20	0.0	Gaussian	Null	4.6	7.1	8.0	2.1	1.0
20	0.0	Gaussian	-3.5%	34.8	40.7	43.6	15.8	12.1
20	0.0	Gaussian	-7%	81.0	84.3	86.2	49.5	46.4
20	0.0	Skewed	Null	11.8	14.6	15.6	19.0	6.3
20	0.0	Skewed	-3.5%	45.4	49.5	51.6	53.8	31.2
20	0.0	Skewed	-7%	77.3	80.3	81.5	82.9	64.7
20	0.0	t_3	Null	5.2	7.5	8.4	2.2	0.9
20	0.0	t_3	-3.5%	43.9	49.8	52.3	32.7	21.2
20	0.0	t_3	-7%	85.7	87.7	88.8	82.8	71.0
20	0.3	Gaussian	Null	10.5	9.3	10.4	4.8	3.1
20	0.3	Gaussian	-3.5%	41.0	37.6	39.9	21.9	17.9
20	0.3	Gaussian	-7%	74.5	70.5	72.6	51.0	46.9
20	0.3	Skewed	Null	17.4	16.5	17.6	20.3	10.3
20	0.3	Skewed	-3.5%	47.8	44.6	46.3	50.9	33.3
20	0.3	Skewed	-7%	75.2	72.7	74.1	77.2	62.1
20	0.3	t_3	Null	11.3	10.0	11.0	5.5	3.1
20	0.3	t_3	-3.5%	48.8	44.7	46.9	35.2	25.9
20	0.3	t_3	-7%	81.3	78.0	79.7	75.4	64.6
20	0.6	Gaussian	Null	20.8	15.9	17.1	11.8	9.3
20	0.6	Gaussian	-3.5%	44.4	36.0	37.7	28.7	24.2
20	0.6	Gaussian	-7%	70.2	62.0	63.8	52.5	47.9
20	0.6	Skewed	Null	27.0	22.1	23.0	24.1	16.9
20	0.6	Skewed	-3.5%	51.7	44.9	46.5	47.2	36.5
20	0.6	Skewed	-7%	71.8	65.3	66.7	68.5	57.1
20	0.6	t_3	Null	21.5	16.0	17.2	12.6	9.4
20	0.6	t_3	-3.5%	49.2	40.8	42.4	36.9	29.6
20	0.6	t_3	-7%	76.2	68.8	70.4	66.9	58.6
33	0.0	Gaussian	Null	4.8	6.0	7.1	3.6	0.8
33	0.0	Gaussian	-3.5%	49.6	53.2	56.3	33.8	18.1
33	0.0	Gaussian	-7%	95.0	95.6	96.1	81.4	68.5
33	0.0	Skewed	Null	10.3	12.0	13.3	39.3	5.9
33	0.0	Skewed	-3.5%	55.5	58.1	60.6	84.7	42.7
33	0.0	Skewed	-7%	89.0	89.8	90.8	98.4	81.8
33	0.0	t_3	Null	5.0	6.5	7.8	4.0	0.7
33	0.0	t_3	-3.5%	58.2	61.0	63.6	62.0	33.2
33	0.0	t_3	-7%	94.4	94.9	95.6	98.3	87.9
33	0.3	Gaussian	Null	11.8	9.1	9.9	8.2	3.0
33	0.3	Gaussian	-3.5%	50.5	44.1	45.3	37.0	23.0
33	0.3	Gaussian	-7%	89.2	85.1	85.5	77.5	63.8
33	0.3	Skewed	Null	17.1	14.4	14.9	36.5	9.4
33	0.3	Skewed	-3.5%	55.7	50.7	51.6	76.8	40.7
33	0.3	Skewed	-7%	85.2	81.8	82.3	95.2	74.7
33	0.3	t_3	Null	11.7	9.3	10.0	9.1	3.3

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Table S17 continued

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
33	0.3	t_3	-3.5%	57.8	51.2	52.2	57.5	33.7
33	0.3	t_3	-7%	90.7	87.9	88.2	94.3	79.5
33	0.6	Gaussian	Null	21.2	14.4	14.1	15.9	8.5
33	0.6	Gaussian	-3.5%	50.8	40.2	39.4	41.1	27.8
33	0.6	Gaussian	-7%	80.7	72.4	71.4	71.5	59.2
33	0.6	Skewed	Null	26.1	20.0	19.5	34.3	14.9
33	0.6	Skewed	-3.5%	55.5	46.7	45.9	65.2	39.0
33	0.6	Skewed	-7%	79.5	72.0	71.2	85.6	65.1
33	0.6	t_3	Null	21.6	14.4	14.1	16.5	8.1
33	0.6	t_3	-3.5%	56.9	46.7	45.8	52.6	34.8
33	0.6	t_3	-7%	84.8	77.6	76.8	84.6	68.4
60	0.0	Gaussian	Null	5.0	5.8	6.5	4.9	0.8
60	0.0	Gaussian	-3.5%	71.8	73.6	74.7	54.4	31.7
60	0.0	Gaussian	-7%	99.7	99.7	99.7	96.8	90.7
60	0.0	Skewed	Null	9.2	10.0	10.6	62.6	5.2
60	0.0	Skewed	-3.5%	70.6	71.7	72.7	98.3	59.5
60	0.0	Skewed	-7%	97.6	97.7	97.9	100.0	95.5
60	0.0	t_3	Null	4.7	5.5	6.2	4.7	0.5
60	0.0	t_3	-3.5%	77.5	78.2	79.1	87.0	57.0
60	0.0	t_3	-7%	98.8	98.7	98.8	100.0	97.6
60	0.3	Gaussian	Null	11.2	8.2	8.3	8.9	2.8
60	0.3	Gaussian	-3.5%	67.5	59.5	59.4	54.2	33.7
60	0.3	Gaussian	-7%	98.4	97.0	96.9	93.9	86.0
60	0.3	Skewed	Null	15.2	11.9	12.1	52.5	8.3
60	0.3	Skewed	-3.5%	67.5	61.3	61.1	93.7	52.2
60	0.3	Skewed	-7%	95.5	93.2	93.1	99.7	89.6
60	0.3	t_3	Null	11.8	8.1	8.1	10.0	2.4
60	0.3	t_3	-3.5%	71.9	64.7	64.5	77.5	47.9
60	0.3	t_3	-7%	97.4	96.5	96.3	99.6	93.9
60	0.6	Gaussian	Null	21.2	13.1	12.3	16.8	7.3
60	0.6	Gaussian	-3.5%	63.3	50.9	48.8	53.3	35.7
60	0.6	Gaussian	-7%	92.7	87.0	85.7	87.2	75.9
60	0.6	Skewed	Null	24.4	17.9	16.9	42.5	13.4
60	0.6	Skewed	-3.5%	64.4	54.2	52.3	80.7	46.7
60	0.6	Skewed	-7%	90.1	84.3	83.2	96.5	79.1
60	0.6	t_3	Null	21.6	13.6	12.5	17.7	7.4
60	0.6	t_3	-3.5%	66.9	56.5	54.2	66.6	43.3
60	0.6	t_3	-7%	92.9	88.4	87.5	95.5	83.1
120	0.0	Gaussian	Null	5.1	5.5	5.9	4.0	0.7
120	0.0	Gaussian	-3.5%	93.3	93.6	93.8	78.8	65.7
120	0.0	Gaussian	-7%	100.0	100.0	100.0	100.0	99.8
120	0.0	Skewed	Null	7.8	8.0	8.4	86.3	4.2
120	0.0	Skewed	-3.5%	90.3	90.5	90.8	100.0	83.8
120	0.0	Skewed	-7%	100.0	99.9	99.9	100.0	99.9
120	0.0	t_3	Null	4.9	5.2	5.6	4.2	0.6

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Table S17 continued

R	ρ	Innovation	Effect	Student	Lag 2	Growing	Sign	Three-test
120	0.0	t_3	-3.5%	93.5	93.6	93.9	98.8	87.1
120	0.0	t_3	-7%	99.7	99.8	99.8	100.0	99.6
120	0.3	Gaussian	Null	11.0	7.9	7.4	7.9	2.9
120	0.3	Gaussian	-3.5%	87.2	81.7	80.2	74.4	60.5
120	0.3	Gaussian	-7%	100.0	100.0	100.0	99.6	99.2
120	0.3	Skewed	Null	14.8	11.0	10.4	71.9	7.4
120	0.3	Skewed	-3.5%	84.6	79.2	77.9	99.7	71.6
120	0.3	Skewed	-7%	99.6	99.2	99.2	100.0	98.6
120	0.3	t_3	Null	12.0	7.9	7.4	8.8	2.3
120	0.3	t_3	-3.5%	88.8	84.5	83.4	94.2	75.5
120	0.3	t_3	-7%	99.7	99.4	99.4	100.0	99.1
120	0.6	Gaussian	Null	20.4	12.5	10.2	15.9	7.2
120	0.6	Gaussian	-3.5%	78.9	67.5	62.8	68.8	53.9
120	0.6	Gaussian	-7%	99.2	98.2	97.2	97.7	95.0
120	0.6	Skewed	Null	23.4	15.9	13.4	53.8	11.9
120	0.6	Skewed	-3.5%	77.5	67.9	63.2	94.3	60.0
120	0.6	Skewed	-7%	97.9	95.6	94.3	99.9	93.2
120	0.6	t_3	Null	21.9	13.5	10.9	17.3	7.2
120	0.6	t_3	-3.5%	81.0	71.7	67.3	83.3	62.1
120	0.6	t_3	-7%	98.7	97.4	96.7	99.8	96.0

S11.3 Cross-record dependence stress test

Proposition 1 requires joint invariance under independent rotations of all monthly records. Marginal cyclic invariance of each record does not imply this product condition. We examined the departure using the 121 observed site coordinates, the five fixed Gaussian graphs and six records from two simulated summers. The June, July and August record lengths were 30, 31 and 31 days in each summer.

For summer y , let

$$U_y(u) = \sum_{k=1}^3 \sqrt{w_k} \{A_{yk} \cos(2\pi ku) + B_{yk} \sin(2\pi ku)\}, \quad u \in [0, 1),$$

where the coefficients are independent standard normal variables and $(w_1, w_2, w_3) = (1, 0.45, 0.20)/1.65$. The same coefficients were used for all three records in a summer. For month m with length n_m , the centred spatial field at $t = 0, \dots, n_m - 1$ was

$$\mathbf{z}_{ymt} = \lambda U_y(t/n_m) \mathbf{g} + \sqrt{1 - \lambda^2} \boldsymbol{\varepsilon}_{ymt}, \quad \lambda \in \{0, 0.3, 0.6, 0.8\}.$$

The vector $\mathbf{g}$ is the centred, unit-mean-square pattern proportional to $0.8x + y$ in projected coordinates. The month-specific Gaussian noise used the first 12 square-root-eigenvalue-weighted eigenmodes of the centred spatial covariance $0.95 \exp(-d/450) + 0.05I$, where d is in km. The retained basis was rescaled to mean site variance one. Its circular moving-average filter had weights $(0.20, 0.50, 1, 0.50, 0.20)$, normalised to unit Euclidean norm. Regional-mean series X_{ymt} were generated from an independent random-number stream with a circular filter proportional to $(0.25, 0.60, 1, 0.60, 0.25)$. We set $\mathbf{y}_{ymt} = X_{ymt} \mathbf{1} + \mathbf{z}_{ymt}$ and used within-record type-7 quartiles of X_{ymt} for the high and middle labels.

Each individual record has a cyclically stationary Gaussian distribution. The shared Fourier coefficients align the three monthly phases within a summer, so independent month-wise rotations alter their cross-covariance when $\lambda > 0$. A numerical covariance check gave a maximum marginal one-step rotation error of 2.0×10^{-15} . Independently rotating the 30-day record by one day changed its cross-covariance with a 31-day record by 0.275 in relative Frobenius norm; the corresponding graph-dispersion cross-covariance changed by 0.362. The pooled same-summer July–August correlation of corresponding daily mean-dispersion values, averaged over the five graphs, increased from 0.003 at $\lambda = 0$ to 0.769 at $\lambda = 0.8$.

For each value of λ , we generated 2,000 replications and used 999 independent product shifts per replication. The statistic averaged the record-specific high-to-middle relative contrasts (ratios minus one) equally over the six records and five graphs. Table S18 reports lower-tail rejection at $\alpha = 0.05$; the seed was 20260812.

Rejection remained between 4.90% and 5.15% as the shared component increased. This result indicates limited sensitivity to the simulated departures. It does not extend Proposition 1 beyond its joint product-invariance condition.

Table S18: Product-shift size under shared cross-record temporal dependence.

Shared loading λ	Rejection rate	Monte Carlo SE
0.0	0.0515	0.00494
0.3	0.0510	0.00492
0.6	0.0495	0.00485
0.8	0.0490	0.00483

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